

# Detect, Refine, Vote: A Calibrated Ensemble for Instance Segmentation of Solar Filaments in GONG $H\alpha$ Images

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**Abstract**—Solar filaments are at the core of many eruptions that drive space weather, but segmenting them as individual instances in full-disk  $H\alpha$  images is hard: they are thin, fade at their ends, and annotators disagree about which faint structures to mark. For the IEEE BigData Cup 2026 Solar Filament Segmentation Challenge (MAGFiLO data, panoptic quality), we combine four instance detectors, an ImageNet-pretrained crop refiner supervised with filament spines, and a native-resolution semantic U-Net in a calibrated ensemble, and evaluate every change out of fold on five random calibration splits. Three lessons emerge. *Vote, don’t select*: replacing the choice of one mask per filament by a score-weighted pixel vote over all overlapping candidates repairs truncated near misses and raises the coverage of thin structures such as barbs from 44% to 49% of their pixels in detected filaments, whereas doubling the refiner’s resolution does not (44.5%). *Separate who may detect from who may vote*: weak or redundant sources lower PQ when they create detections but raise it as voters; with the semantic model as an extra detector inside the vote, PQ rises from 0.450 to 0.461 (positive on all five splits), and the pipeline scores 0.40 on the public leaderboard without using test annotations. *The headroom is small*: two annotators of the same image agree at PQ 0.31–0.34, below the model’s 0.456 against those annotators.

**Index Terms**—solar filaments, instance segmentation, panoptic quality, ensemble learning, H-alpha, space weather

## I. INTRODUCTION

Filaments are dense, cool plasma held by magnetic fields above polarity inversion lines; they appear as dark, elongated features on the  $H\alpha$  disk and are at the core of coronal mass ejections and flares. Instance-level segmentation of filaments in the continuous full-disk  $H\alpha$  stream of the Global Oscillation Network Group (GONG) [1] is therefore useful for space-weather research. The MAGFiLO dataset [2] provides manually annotated filament polygons and spines for GONG images, and the IEEE BigData Cup 2026 challenge built on it scores instance segmentations by panoptic quality (PQ) [3].

Three properties make the task harder than generic instance segmentation: filaments are thin (a few pixels wide in a  $2048 \times 2048$  image) and fade gradually at their ends; neighbouring filaments touch or fragment; and labels are noisy, since 42% of the training images were annotated by two or three people who often disagree on which structures are filaments.

**Related work.** Deep networks have been applied to filament segmentation in  $H\alpha$  images by Ahmadzadeh et al. [4], Zhu et al. [5] and Guo et al. [6]; Diercke et al. [7] combine object

detection with semantic segmentation in a semi-supervised setting. We follow the detect-then-segment idea but build it as a calibrated ensemble scored at the instance level, where detection, mask quality and the separation of touching filaments all enter the metric.

**Contributions.** (i) A detect–refine–vote pipeline whose final stage replaces “keep the best-scoring mask” by a score-weighted pixel vote over overlapping candidates, with the roles of *leader* (may create a detection) and *voter* (only shapes a detection) assigned separately (Algorithm 1). (ii) Evidence for three lessons: voting repairs masks, including thin structures and near misses, where higher refiner resolution does not (Table III); sources that hurt as detectors help as voters (Table II, Fig. 3); and human agreement bounds the reachable PQ. (iii) An out-of-fold evaluation protocol with paired bootstrap, under which most of the roughly 50 variants we tried were rejected (Table IV), reported together with the organizers’ overlap statistics.

## II. DATA AND EVALUATION PROTOCOL

**Data.** The challenge training set has 707 GONG  $H\alpha$  images ( $2048 \times 2048$  pixels, six observatories, 2011–2022) with 1,154 annotation sets (an image annotated by  $k$  people appears  $k$  times) and 8,199 filament polygons with spine polylines. The test set has 180 images; the public leaderboard uses about half of them. We used only the challenge’s training annotations; no MAGFiLO test annotations were used.

**Split and metric.** We hold out 10% of the *images* (70 images, 116 annotation sets, 908 filaments), so that no image is shared with training through a second annotator, and score each annotation set separately.  $PQ = \sum_{TP} \text{IoU} / (|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|)$  with matches at  $\text{IoU} > 0.5$ , pooled over all images, follows the organizers’ self-evaluation notebook; we also report segmentation quality (SQ, mean IoU of matches) and recognition quality ( $RQ = PQ/SQ$ ).

**Out-of-fold protocol.** Calibration models are fitted on validation candidates, so in-sample scores are optimistic. For every comparison we split the 70 validation images into two folds, fit calibrators on one and score the other, repeat this for five random fold assignments (the same five for all rows), and compute a 2,000-sample paired bootstrap over images of the PQ difference. These five splits share the same held-out

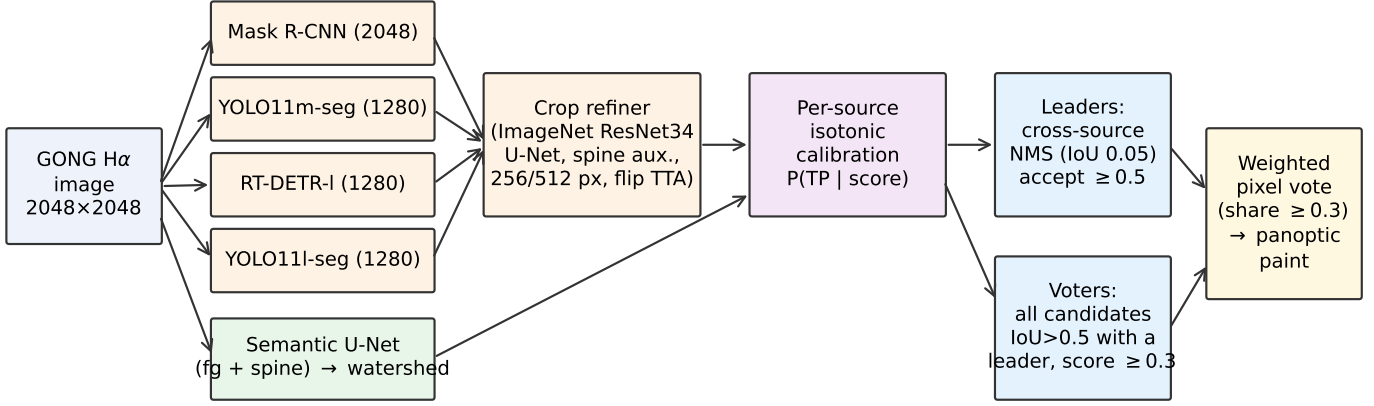


Fig. 1. Detect-refine-vote pipeline. Detector proposals are re-segmented by a crop refiner, scores are calibrated per source, cross-source NMS selects leaders, and each leader’s final mask is a score-weighted pixel vote over all overlapping candidates. The semantic U-Net’s instances act as additional leaders and voters.

images and the same trained models, so they are correlated: they control the calibration step, not training randomness or the choice of held-out images. Retraining one detector with another seed changed ensemble PQ by up to  $\pm 0.005$ , which is the scale below which we do not interpret differences. Once this protocol was in place, we kept a change only if it improved PQ on all five splits.

### III. METHOD

Fig. 1 summarizes the pipeline; code is available at <https://github.com/biconcavelens/filament-segmentation-2026>.

#### A. Instance detectors

We train four detectors with different architectures and input scales: Mask R-CNN [8] (torchvision ResNet-50-FPN v2, COCO-initialized) at the native 2048 px, YOLO11m-seg and YOLO11l-seg [9] at 1280 px, and RT-DETR-I [10] at 1280 px. Well-localized boxes often received very low confidence, so we up-weighted the classification loss ( $3\times$  for Mask R-CNN,  $cls=1.5$  for YOLO,  $cls=2.0$  for RT-DETR); for RT-DETR this raised the median score of boxes with  $IoU > 0.5$  from 0.03 to 0.66. Training Mask R-CNN at native resolution instead of 1333 px added +0.0035 PQ to a two-detector ensemble (all five splits).

#### B. Crop refiner

Detector mask heads are coarse, so every proposal is re-segmented by a refiner: a square crop of  $1.8\times$  the proposal’s extent is resized to  $256 \times 256$  and passed to a U-Net [11] with an ImageNet-pretrained ResNet-34 encoder [12], [13]. Besides the mask, the refiner predicts the annotated spine (centerline) as an auxiliary output. It is trained on crops around ground-truth instances with BCE+Dice (weight 0.7) and spine BCE (0.3), and applied with four-view flip test-time augmentation. Replacing a from-scratch U-Net of similar size by the pretrained encoder lowered validation loss from 0.103 to 0.092 and raised ensemble PQ by +0.006 to +0.012 on all

five splits. A second refiner at  $512 \times 512$  is used only as a voter.

#### C. Semantic model

A U-Net with an ImageNet ConvNeXt-Tiny encoder [14] is trained at native resolution on  $1024 \times 1024$  crops, with focal [15] plus Dice [16] loss on the foreground and BCE plus Dice on a 5 px rasterized spine (dihedral and on-disk photometric augmentation, AdamW, cosine schedule, 62 epochs in 3 h on one Kaggle GPU). Instances are obtained by thresholding the foreground at 0.5, using connected components of the spine map ( $> 0.3$ ) as watershed seeds to split touching filaments, and dropping regions under 100 px; an instance’s score is its mean foreground probability.

#### D. Calibration, leaders and cluster voting

Raw scores are not comparable across sources (Fig. 2): a calibrated probability of 0.5 corresponds to a raw score of 0.38 for YOLO11m-seg but 0.65 for RT-DETR and 0.89 for Mask R-CNN. Each source’s scores are therefore mapped to  $P(TP | score)$  by isotonic regression [17] fitted on validation candidates, where a candidate is a TP if its refined mask has  $IoU > 0.5$  with an annotated filament.

Algorithm 1 gives the selection and voting step. Calibrated candidates of the leader sources are pooled, and a greedy NMS keeps those with  $P \geq 0.5$  that overlap no kept candidate by more than  $IoU 0.05$ ; these are the leaders. Instead of keeping only each leader’s own mask, we fuse it with the masks that describe the same filament, in the spirit of weighted box fusion [18]: every other candidate with  $P \geq 0.3$  joins the leader it overlaps most if that  $IoU$  exceeds 0.5, and the leader’s final mask keeps the pixels whose score-weighted share of its cluster is at least 0.3. Leaders come from the four detectors (256 px refiner) and the semantic instances; voters additionally include the four detectors refined at 512 px and run on the horizontally mirrored image. Masks are finally painted in score order so that predictions do not overlap.

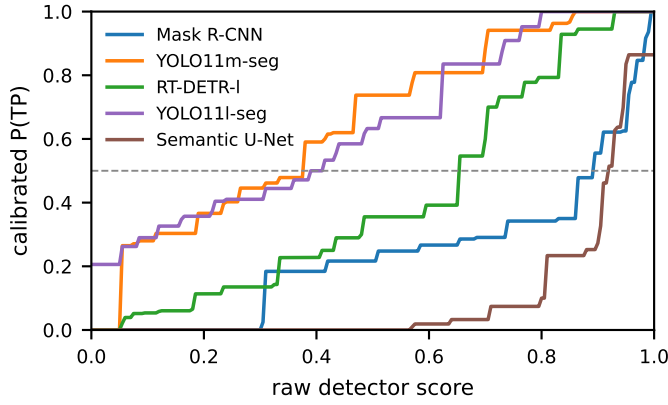


Fig. 2. Isotonic calibration of each source’s raw score into  $P(TP)$  on validation candidates. The dashed line marks the acceptance threshold used for leaders.

#### Algorithm 1 Leader selection and cluster voting (per image)

**Require:** candidates  $(m_i, p_i, s_i)$ : mask, calibrated score, source; leader sources  $\mathcal{L}$

- 1:  $L \leftarrow \emptyset$
- 2: **for**  $i$  in order of decreasing  $p_i$  **do**
- 3:   **if**  $s_i \in \mathcal{L}, p_i \geq 0.5, \text{IoU}(m_i, m_j) \leq 0.05 \forall j \in L$  **then**
- 4:      $L \leftarrow L \cup \{i\}$
- 5:   **end if**
- 6: **end for**
- 7:  $C_\ell \leftarrow \{\ell\}$  for each  $\ell \in L$
- 8: **for** each  $j \notin L$  with  $p_j \geq 0.3$  **do**
- 9:    $\ell^* \leftarrow \arg \max_{\ell \in L} \text{IoU}(m_j, m_\ell)$
- 10:   **if**  $\text{IoU}(m_j, m_{\ell^*}) > 0.5$  **then**  $C_{\ell^*} \leftarrow C_{\ell^*} \cup \{j\}$
- 11:   **end if**
- 12: **end for**
- 13:  $\hat{m}_\ell(x) \leftarrow [\sum_{k \in C_\ell} p_k m_k(x) / \sum_{k \in C_\ell} p_k \geq 0.3]$
- 14: **end for**
- 15: paint  $\hat{m}_\ell$  in order of decreasing  $p_\ell$  onto unclaimed pixels; drop fragments  $< 20$  px

## IV. EXPERIMENTS

**Main results.** Table II and Fig. 3 list single sources and pipeline stages. Each detector alone reaches PQ 0.42–0.44; the calibrated ensemble of four reaches 0.450, mainly through recognition (higher RQ at equal SQ). Adding the semantic instances as plain detections does not help (−0.001 on average, −0.006 to +0.003 over splits), but when they and the extra refinements only vote, or lead through the vote, PQ rises to 0.461 and matched filaments increase from 568 to 600 of 908 (means over splits). The final configuration improves on the baseline on all five splits, and its 95% bootstrap interval excludes zero on three of them; with 70 validation images this is consistent but modest evidence. On the public leaderboard, a single Mask R-CNN scored 0.27, adding the crop refiner 0.34, the calibrated two-detector ensemble 0.39, and the pretrained refiner 0.40; all voting variants also score 0.40. With about 90

TABLE I  
TRAINING SETUP (EACH MODEL ON A SINGLE 16 GB KAGGLE GPU OR AN 8 GB LAPTOP RTX 4060).

| Model          | Init., input                   | Training                                                                            |
|----------------|--------------------------------|-------------------------------------------------------------------------------------|
| Mask R-CNN     | COCO, 2048                     | SGD lr $2.5 \times 10^{-3}$ , 6 epochs, batch 1, cls. loss $\times 3$ , flips       |
| YOLO11m-seg    | COCO, 1280                     | 40 epochs, batch 4, cls= 1.5                                                        |
| YOLO11l-seg    | COCO, 1280                     | 103 epochs (time budget), cls= 1.5                                                  |
| RT-DETR-l      | COCO, 1280                     | 60 epochs, batch 2, cls= 2.0                                                        |
| Crop refiner   | ImageNet R34, 256 crop         | Adam $3 \times 10^{-4}$ , cosine, 15 epochs, batch 16, rot./flip/brightness         |
| Semantic U-Net | ImageNet ConvNeXt-T, 1024 crop | AdamW $3 \times 10^{-4}$ , cosine, 62 epochs (3 h), batch 4, dihedral + photometric |

TABLE II  
VALIDATION RESULTS, OUT OF FOLD: MEAN PQ (STANDARD DEVIATION) OVER FIVE RANDOM CALIBRATION SPLITS, SQ AND RQ, AND THE PQ CHANGE VERSUS THE CALIBRATED 4-DETECTOR ENSEMBLE (RANGE OVER SPLITS; NUMBER OF SPLITS WHOSE 95% BOOTSTRAP INTERVAL EXCLUDES ZERO). ROWS STARTING WITH + MODIFY THE 4-DETECTOR ENSEMBLE. LAST COLUMN: PUBLIC LEADERBOARD.

| Configuration            | PQ          | SQ   | RQ   | $\Delta$ PQ range (sig.) | Pub. |
|--------------------------|-------------|------|------|--------------------------|------|
| Mask R-CNN               | .429 (.004) | .682 | .629 |                          |      |
| YOLO11m-seg              | .437 (.002) | .684 | .639 |                          |      |
| RT-DETR-l                | .436 (.002) | .686 | .636 |                          |      |
| YOLO11l-seg              | .420 (.004) | .687 | .611 |                          |      |
| Semantic U-Net           | .396 (.003) | .669 | .591 |                          |      |
| 4 detectors + NMS        | .450 (.003) | .682 | .660 | —                        | .40  |
| + semantic, no vote      | .448 (.001) | .678 | .661 | [−.006, +.003] (0/5)     |      |
| + vote, 4 voters         | .453 (.003) | .682 | .664 | [+.003, +.004] (0/5)     | .40  |
| + vote, 12 voters        | .457 (.002) | .684 | .668 | [+.005, +.009] (3/5)     | .40  |
| + sem. leader, 5 voters  | .455 (.001) | .680 | .668 | [+.001, +.009] (0/5)     |      |
| + sem. leader, 13 voters | .461 (.002) | .683 | .675 | [+.007, +.016] (3/5)     | .40  |

public images and two-decimal reporting, gains of  $< 0.01$  are not resolvable there, and our validation scores run about 0.05 above the public ones for every configuration.

**What voting and the semantic leaders change.** Table III measures fine structures, the thin parts of annotated filaments that include barbs and faint ends. In the 4-detector ensemble, detected filaments have 44% of their thin pixels covered against 81% of their core, and refining at 512 instead of 256 px barely changes this (44.5%). Voting raises it to 49% (core 84%) without lowering SQ, so the gain is not simply larger masks; uniformly dilating masks by 1 px lowered PQ from 0.446 to 0.419 in an earlier version of the pipeline. The ordering is the same with a radius of 3 px (thin coverage  $0.58 \rightarrow 0.63$ ). Voting alone creates no new detections: with 12 voters it converts 13 of the 96 near misses into matches and loses 4 matches. The semantic leaders add 30 filaments that every baseline prediction missed (best IoU  $< 0.3$ ), which accounts for most of the recall gain. The semantic model alone draws the thin parts of the filaments it detects as completely as the full vote (50%) but finds far fewer filaments, which is why it works best as a leader inside the vote.

**Overlap statistics.** The organizers additionally evaluate the distributions of IoU and Dice over all overlapping prediction–annotation pairs and the one-to-many and many-to-one rela-

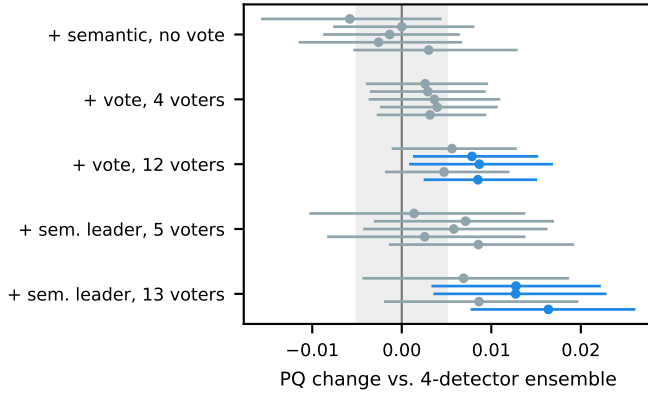


Fig. 3. PQ change versus the calibrated 4-detector ensemble on each of the five calibration splits (points) with paired-bootstrap 95% intervals; blue intervals exclude zero. The gray band marks  $\pm 0.005$ , the change caused by retraining one detector with another seed.

TABLE III

FINE STRUCTURES AND NEAR MISSES (ONE OUT-OF-FOLD SPLIT). THIN: PIXELS OF EACH ANNOTATED FILAMENT REMOVED BY A MORPHOLOGICAL OPENING WITH A DISK OF RADIUS 2 PX (PARTS NARROWER THAN  $\approx 5$  PX; 1.9% OF ANNOTATED PIXELS); CORE: THE REST. COVERAGE OF THIN AND CORE PIXELS BY ANY PREDICTION, AND BY THE MATCHED PREDICTION OF DETECTED FILAMENTS ( $\text{IoU} > 0.5$ ). NEAR MISS: ANNOTATED FILAMENTS WITH BEST  $\text{IoU}$  0.3–0.5 IN THE 4-DETECTOR ENSEMBLE (96) THAT BECOME MATCHES.

| Configuration               | Any prediction |      | Matched |      | Near misses |
|-----------------------------|----------------|------|---------|------|-------------|
|                             | Thin           | Core | Thin    | Core |             |
| 4 detectors (256px refiner) | .361           | .637 | .441    | .813 | 560         |
| 4 detectors (512px refiner) | .364           | .638 | .445    | .815 | 562         |
| + vote, 4 voters            | .395           | .656 | .478    | .832 | 564         |
| + vote, 12 voters           | .407           | .661 | .491    | .840 | 569         |
| + sem. leader, 13 voters    | .432           | .694 | .502    | .841 | 600         |
| Semantic U-Net alone        | .356           | .569 | .498    | .845 | 462         |

tions, computed as in their self-evaluation notebook (Fig. 4). For the final pipeline the mean  $\text{IoU}$  over overlapping pairs is 0.614 versus 0.608 for the 4-detector ensemble (mean Dice 0.744 vs. 0.739). Annotated filaments with no overlapping prediction fall from 212 to 181 of 908, at the cost of more predictions without any overlapping annotation (126 vs. 104). Fragmentation and merging are rare for both: 19 vs. 15 annotated filaments overlap two or more predictions (one-to-many) and 14 vs. 16 predictions overlap two or more annotations (many-to-one).

**Where the error is.** For the 4-detector ensemble (one calibration split), 325 of 908 validation filaments are missed. For 219 of them a correct mask exists among the candidates but its calibrated score is below the acceptance threshold (median 0.33); 67 have no correct candidate (57 of these have a near miss at  $\text{IoU}$  0.3–0.5, typically a mask truncated before the filament’s faint tail); 39 lose to an overlapping wrong candidate. Lowering the threshold does not pay off: leaders scored 0.3–0.5 are true positives at most 32% of the time even when nine or more sources agree, below the  $\approx 0.34$  rate at which accepting a detection increases PQ.

**Human agreement.** On the 296 training images annotated

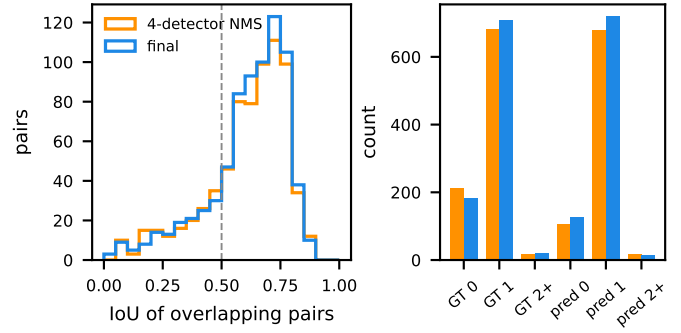


Fig. 4. Organizers’ overlap statistics on validation (one out-of-fold split). Left:  $\text{IoU}$  of every overlapping prediction–annotation pair. Right: number of annotated filaments (GT) and predictions overlapped by 0, 1, or  $\geq 2$  partners; “GT 2+” is fragmentation (one-to-many), “pred 2+” merging (many-to-one).

TABLE IV

REJECTED ALTERNATIVES. CONTEXT: THE SYSTEM THE CHANGE WAS TESTED IN (Y = YOLO11M-SEG ALONE; E = 4-DETECTOR ENSEMBLE WITH THE EARLIER FROM-SCRATCH REFINER; F = 4-DETECTOR ENSEMBLE WITH THE PRETRAINED REFINER, WITH OR WITHOUT VOTING). PQ ON VALIDATION; OOF RANGES OVER FIVE SPLITS WHERE MARKED.

| Alternative                                                                | Ctx.  | Effect on PQ                   |
|----------------------------------------------------------------------------|-------|--------------------------------|
| Mask2Former (Swin-T), frozen or fine-tuned encoder                         | E     | solo $\leq .345$ ; lowers E    |
| SAM ViT-B decoder, box prompts                                             | –     | solo .316                      |
| SimSiam pretraining of the Mask R-CNN backbone on 4k unlabeled GONG images | E     | tie (.448 vs .449); public .38 |
| Self-training on pseudo-labeled test images                                | Y     | .417 $\rightarrow$ .393        |
| CLAHE contrast enhancement                                                 | Y     | .425 $\rightarrow$ .412–.419   |
| Gradient-boosted rescorer (agreement, shape features)                      | F     | –.014 to –.004 (OOF)           |
| Growing masks into U-Net foreground                                        | E / F | + .002 / –.006                 |
| Mirrored-image proposals as detections                                     | F     | –.009 to +.003 (OOF)           |
| Second-seed or augmented Mask R-CNN                                        | F     | –.005 to +.003 (OOF)           |
| Two semantic models averaged                                               | F     | –.006 to –.003 (OOF)           |

by two or more people, scoring one annotator against another with the challenge metric gives PQ 0.340 (SQ 0.634, RQ 0.536). On the 33 validation images with several annotators the human–human PQ is 0.309, while our model scores 0.456 (the 4-detector ensemble 0.447) out of fold against the same annotation sets. Trained on all annotators, the model approximates their consensus and agrees with any single annotator better than annotators agree with each other; much of the remaining “error” is disagreement about which faint structures are filaments and where their tails end.

**What did not help.** Table IV lists the main rejected alternatives. Two patterns recur. Adding *detections* from further correlated sources (a second-seed Mask R-CNN, mirrored-image proposals, the semantic instances without voting) adds near-duplicate false positives faster than true positives, while the same sources help as voters. Learned rescoring with richer features improved candidate-level ranking but not PQ.

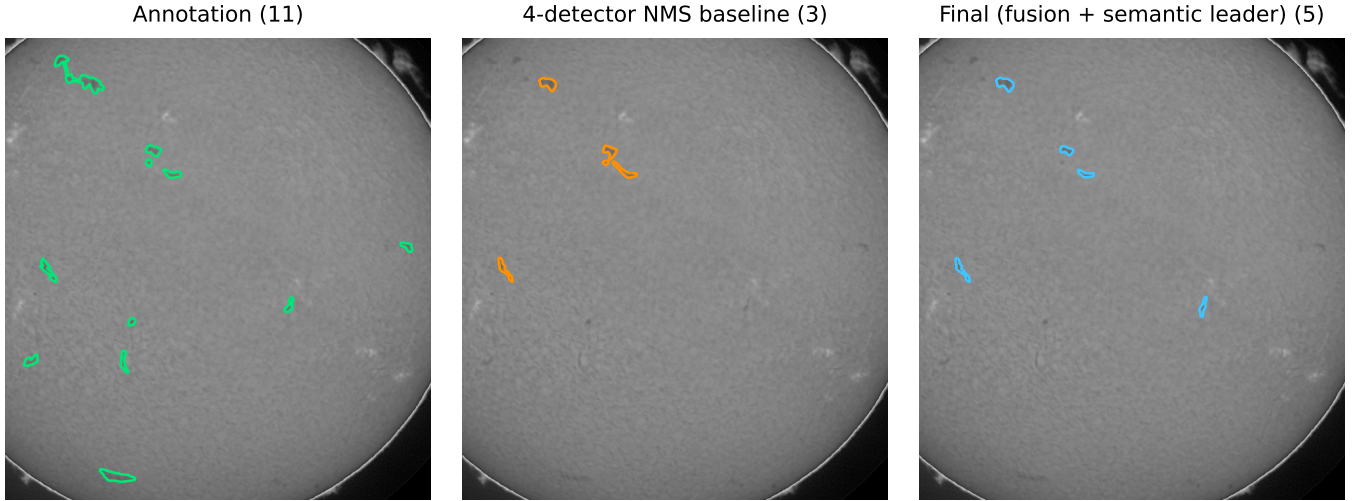


Fig. 5. Validation example where voting helps (chosen as the image with the largest gain in matched filaments, not as a typical case). Left: one annotator’s filaments. Middle: the 4-detector ensemble merges two filaments into one instance and misses several faint ones. Right: the final pipeline separates the pair and recovers two more; several faint annotated filaments remain undetected.

## V. DISCUSSION

**Cost.** Refinement dominates inference time. Per validation image the four detectors produce about 93 candidates above their score floors (13, 20, 41 and 19), each refined with four-view TTA; on a 16 GB Kaggle GPU the Mask R-CNN source including its refinement takes 1.9–2.2 s per image and the semantic U-Net 1.2 s per image (2048 px, four-view TTA). The full configuration refines every candidate three times (256 px, 512 px, mirrored image). A light configuration that keeps the semantic leader but uses only the five leader sources as voters retains about 45% of the gain (+0.005 PQ) at about a third of the refinement cost.

**Limitations.** The thresholds (acceptance 0.5, NMS 0.05) and fusion parameters were chosen on the same 70 validation images we report on; for the fusion parameters the in-sample PQ varies by only 0.001 over a 27-point grid, so this selection effect is small, but the other choices may be optimistic. The models were trained once on one split, so the five splits do not capture training variance. Fine structures remain the weakest part: even the final pipeline covers about half of the thin pixels of the filaments it detects, against 84% of their core; neither the refiner nor the semantic model is trained to emphasize barbs, which are also inconsistently annotated. The 0.05 gap between validation and public scores is consistent across configurations; we have not identified its cause.

## VI. CONCLUSION

A calibrated ensemble of standard detectors, a pretrained spine-supervised crop refiner, and a cluster-level pixel vote reaches PQ 0.40 on the public leaderboard without test-label leakage. Three lessons carry beyond our pipeline. Voting over overlapping masks instead of selecting one repairs truncated masks and thin structures, which higher refiner resolution does not. Separating the right to detect from the right to vote

lets weak or redundant sources help instead of adding false positives. And because human annotators agree with each other at only PQ 0.31–0.34, progress on this benchmark is best judged with out-of-fold protocols and distributional measures, such as the coverage of thin structures, rather than leaderboard differences of a few thousandths.

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